

OmniRouting: A Semantic-Coupled Multimodal Benchmark for Constraint-Aware Spatial Reasoning in PCB Routing

Anonymous submission

Abstract

Recent large language models (LLMs) have demonstrated remarkable progress in constraint-aware navigation, maze reasoning, graph reasoning. However, their ability to reason about complex routing problems under strict geometric, topological, and electrical constraints remains largely unexplored, despite routing being one of the most challenging and critical stages of electronic design automation (EDA). To bridge this gap, we introduce **OmniRouting**, the first large scale benchmark designed to evaluate LLMs on printed-circuit-board (PCB) routing reasoning under real-world industrial design-rule, manufacturability, and connectivity constraints. **OmniRouting** contains 1,681 industrial-grade schematic-coupled PCB designs, including board geometries, rout-able component placements by human engineer, footprints, pad locations, netlists, stackup information, and routing constraints. The benchmark comprises four tasks: (1) **geometric routing reasoning**, generating physically valid copper traces, vias, and layer assignments to connect circuit nets within constrained board regions; (2) **design-rule-aware routing reasoning**, producing routable layouts that satisfy clearance, trace-width, via, obstacle-avoidance, and board-boundary constraints; (3) **electrical functionality reasoning**, preserving schematic-specified connectivity while reasoning over net names and functional roles to produce electrically correct routing; and (4) **tool-augmented agentic routing**, leveraging external tools for (1)-(3). Our results reveal substantial limitations of current LLMs in PCB routing, including weak path-planning capabilities, poor adherence to design-rule constraints, and inconsistent preservation of electrical functionality. We will open-source all benchmark data, evaluation code, and tool interfaces to facilitate future research.

Introduction

As the scale, integration level, and component density of integrated circuits (ICs) continue to increase, the complexity of their physical realization on printed circuit boards (PCBs) grows accordingly. Since PCBs provide the physical interconnections required to realize system functionality, PCB EDA tools play a critical role in modern electronic design. Among the stages of the PCB design flow, routing is one of the most challenging and computationally intensive combinatorial optimization problems. Given a fixed component placement and connectivity specification, routing generates conductive interconnects between component pins while minimizing wirelength and satisfying man-

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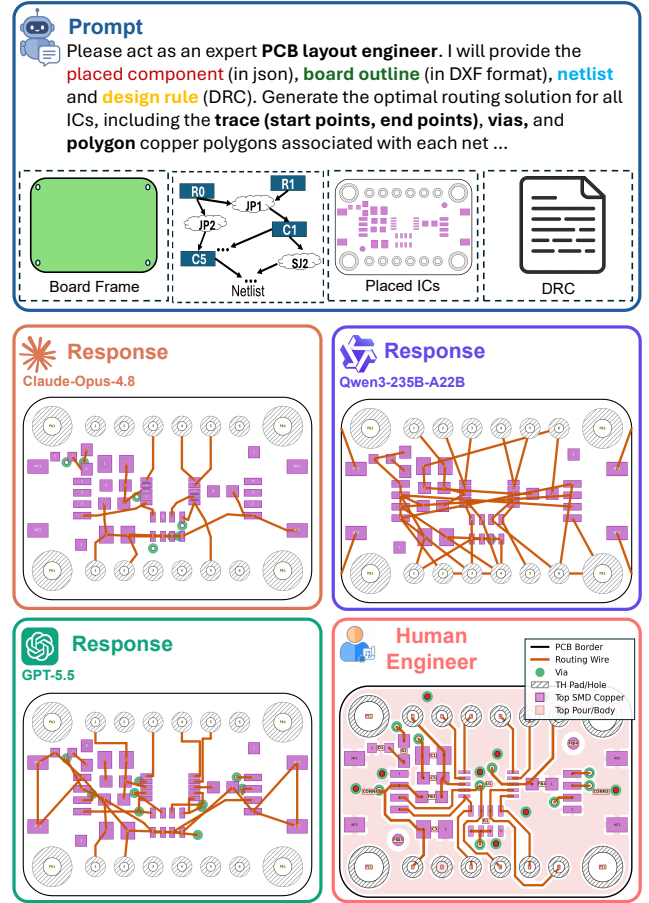


Figure 1: **Large language models (LLMs) fail to perform reliable PCB routing under complex geometric, electrical, and manufacturing constraints.** Frequent routing violations (highlighted by dashed red circles), including unrouted connections, trace conflicts, and constraint violations, reveal substantial weaknesses in spatial reasoning and constraint-aware routing generation.

ufacturing and design constraints. Unlike IC routing, which benefits from highly standardized design abstractions and consequently achieves a high degree of automation, PCB routing remains heavily dependent on expert layout engineers because of its inherently irregular geometries, diverse

component layouts, and complex routing constraints. Additionally, while IC routing research benefits from extensive benchmark suites comprising hundreds or even thousands of designs (Alpert 1998; Nam, Sze, and Yildiz 2008; Mantik et al. 2018), publicly available PCB routing datasets remain scarce. Because industrial PCB layouts are almost exclusively proprietary, most existing PCB routing studies (Xie et al. 2018; Chen et al. 2020; Li et al. 2023) rely on fewer than 20 real-world board designs for training and evaluation.

In recent years, large multimodal models (LMMs) (Chen et al. 2021, 2024a; Yuan, Wang, and Waslander 2025; Dao and Vu 2025), have demonstrated strong capabilities in constraint-aware navigation, visual spatial reasoning, topological planning, and fine-grained geometry. These advances suggest new opportunities for applying LMMs to PCB routing, which can be viewed as a constrained path-planning problem requiring the generation of valid trace paths while satisfying diverse constraints. However, existing spatial-LMM works largely overlook the core difficulty of PCB routing: generating valid trace paths in irregular, congested layouts while satisfying design-specific constraints such as clearance, trace width, layer transitions, differential pairs, length matching and signal integrity. These constraints are difficult to fully formalize and often require expert routing knowledge. This raises a critical open question: *Can current LMMs generate PCB routing solutions that satisfy geometric constraints, preserve electrical connectivity, and remain functional across diverse real-world circuit designs?*

To investigate this question, we conducted a series of preliminary experiments to assess the PCB routing performance of several state-of-the-art LMMs, including Claude-Sonnet-4.6 (Anthropic 2026), GPT-5.5 (OpenAI 2026), and Qwen3.6-35B-A3B (Qwen Team 2026). Their performance was compared against that of experienced human layout engineers. Specifically, we evaluated their ability to generate routing solutions for real-world PCB designs with engineer-crafted component placements that have been verified to be fully routable. This setup isolates the routing task from placement quality and ensures that any failures arise from routing rather than unroutable component arrangements. As illustrated in Figure 1, all evaluated models frequently produce routing failures, including unrouted connections, trace conflicts, and design-rule violations, such as routing GND as ordinary traces instead of implementing continuous ground polygons and return-current paths. These results reveal that current LMMs fundamentally lack the topological planning and constraint-aware navigation capabilities required for PCB routing, frequently failing to generate complete, conflict-free, and electrically functional routing solutions in congested routing environments.

To bridge this gap, we introduce *OmniRouting*, the first large-scale benchmark for evaluating LMMs on schematic-coupled PCB routing reasoning, comprising industrial-grade designs with 100% routable component layouts and complete ground-truth routing implementations, including all connection geometries (trace), layer transitions (via), and conductive structures (polygon) required for physical realization. As shown in Figure 2, *OmniRouting* contains 1,681 real-world PCB layouts paired with corresponding schematics and

expert-designed reference layouts, spanning a wide range of board areas ($xx - xx \text{ mm}^2$) and PCB stackups from 2 to 8 layers. The benchmark comprises 109.9K component placement instances, 245.4K pads and SMDs, 219.8K semantically annotated net connections, XXXK vias, and YYYY copper pours, capturing complete PCB routing implementations, including traces, vias, and plane geometries, that pass industrial design-rule checks (DRC) and satisfy fabrication constraints. We first propose a systematic evaluation protocol for assessing PCB routing designs generated by LMMs with respect to connectivity, routing legality, and electrical functionality. Our benchmark evaluates four core abilities, including (i) **topological path planning** for constructing complete routing topologies that connect circuit nets, (ii) **design-rule-aware routing reasoning** for generating manufacturable routing solutions under geometric and fabrication constraints, (iii) **electrical functionality reasoning** preserving the electrical intent of expert-designed layouts, as measured by routing-path, via-placement, and topology similarity to reference designs; and (iv) **tool-augmented agentic routing** for iterative routing refinement through benchmark-provided routing analysis and optimization tools.

Our main contributions are as follows: (i) We introduce *OmniRouting*, a comprehensive benchmark for evaluating LMMs on PCB routing reasoning, containing 1,681 real-world schematic-coupled PCB designs with expert-designed reference routings, complete ground-truth traces, vias, and copper pours, as well as systematic evaluation protocols for routing completion, connectivity, design-rule compliance, routing quality, and reference similarity. (ii) We systematically evaluate state-of-the-art LMMs on PCB routing under multiple settings, including zero-shot routing generation and tool-augmented agentic routing with iterative visualization, scoring, and routing-refinement feedback. (iii) We provide a detailed analysis of LMM capabilities in PCB routing, examining performance across topological routing reasoning, design-rule-aware routing generation, preservation of electrical intent, routing quality optimization, and the effectiveness of tool-augmented agentic routing through iterative interaction with routing analysis tools.

Related Work

Topological Planning in LMMs. Recent LLMs have demonstrated increasingly strong capabilities in constraint-aware navigation, maze reasoning, and graph reasoning. Early studies such as NavGPT and PPNL explore whether LLMs can follow navigation instructions, plan feasible paths, avoid obstacles, and satisfy spatial-temporal constraints in embodied or grid-world environments (Zhou, Hong, and Wu 2024; Aghzal, Plaku, and Yao 2023). Subsequent works further evaluate maze-style reasoning and spatial state tracking through visualization-based prompting and coordinate-based navigation benchmarks (Wu et al. 2024; Einarsson 2026). In parallel, graph-reasoning benchmarks such as NL-Graph, Talk like a Graph, GraphInstruct, GraphWiz, and GraphArena examine LLMs’ ability to understand graph-structured inputs and solve connectivity, shortest-path, traversal, flow, and combinatorial graph problems (Wang et al. 2023; Fatemi, Halcrow, and Perozzi 2024; Luo et al. 2024;

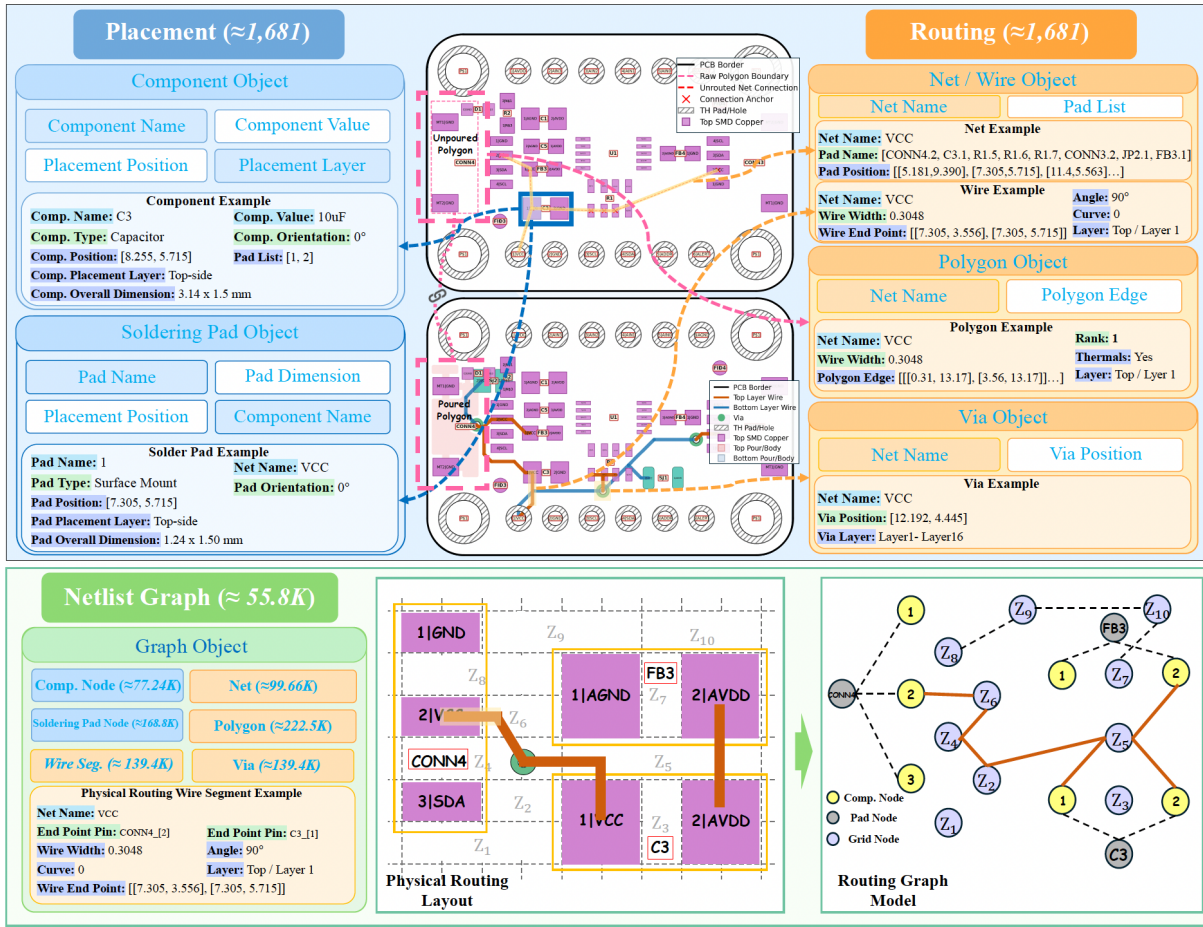


Figure 2: Overview of *OmniRouting* benchmark with representative cases.

Chen et al. 2024b; Tang et al. 2025). Despite these advances, existing LMM research has largely overlooked PCB routing, a substantially more challenging problem than traditional constrained navigation. Rather than planning a single collision-free path, PCB routing requires jointly optimizing thousands of interdependent routes under heterogeneous geometric, electrical, and manufacturability constraints.

Benchmarks for PCB Routing. In recent years, a growing number of both commercial tools (Cadence Design Systems 2026; Eremex, Ltd. 2026; FreeRouting Project 2026; Quilter 2026) and academic methods (Liao et al. 2020; He and Bao 2020; Tseng et al. 2015; Fang et al. 2024) have been proposed to automate the PCB routing process, spanning classical maze routing, rip-up-and-reroute, push-and-shove and topological routers, constraint-based optimization, and emerging learning-based approaches. Representative commercial systems such as Cadence Allegro X AI, Allegro/Specctra, TopoR, FreeRouting/ELECTRA, and Quilter provide automatic or AI-assisted routing capabilities, including critical-net routing, shape-based routing, and physics-driven layout optimization. However, these tools are typically developed and validated within proprietary industrial flows, where the underlying board designs, routing constraints, layer stack-ups, design rules, and generated routing solutions are not publicly released, posing significant barriers to reproducible

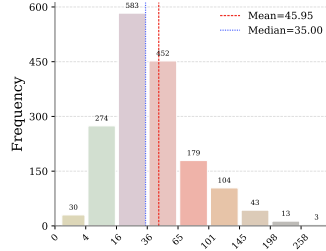
benchmarking and fair comparison. Recent academic efforts such as deep reinforcement learning for global routing (Liao et al. 2020), MCTS-guided neural circuit routing (He and Bao 2020), ILP-based meander-segment alleviation (Tseng et al. 2015), and obstacle-aware length-matching routing (Fang et al. 2024) have demonstrated promising progress in routing formulation, search strategy, and constraint handling. However, existing PCB routing benchmarks remain limited in both scale and realism: publicly available datasets typically contain no more than 20 board designs, with most instances restricted to simple two-layer PCBs and relatively small routing problems. As a result, PCB routing still lacks an open benchmark suite containing realistic multi-layer industrial boards, complex component placements, heterogeneous net classes and clearance constraints.

Benchmark Construction

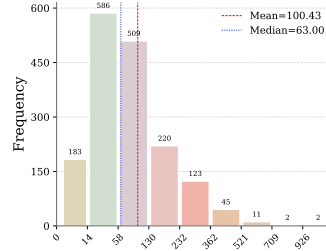
Task Formulation. To provide a comprehensive evaluation framework for PCB routing tasks, our task formulation covers four key capabilities of constraint-aware geometric reasoning: (i) **Connectivity**, which requires generating routing paths that completely connect every net while preserving the intended electrical topology, evaluated by routability, open count, and short count; (ii) **Design-Rule Compliance**,



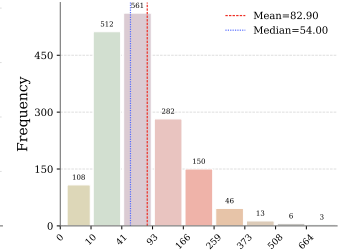
(a) worldcloud of topics



(b) Components per diagram.



(c) Pins per diagram.



(d) Nets per diagram.

Figure 3: Statistical overview of the *OmniRouting* benchmark. The dataset covers a wide range of electronic domains, with each layout design containing up to **327 components**, **1,172 pins**, and **840 nets**.

which requires producing manufacturable routing that satisfies physical design constraints, evaluated by total DRC, clearance, trace-width, via-rule, and boundary/keepout violations; (iii) **Electrical Functionality**, (iv) **Agentic Tool Use for Constraint-Aware Geometric Reasoning**, which formulates PCB routing as an iterative decision-making problem, where LMMs are equipped with tools (such as routing visualization, conflict checking, and connectivity verification) to progressively refine routing paths until all geometric and electrical constraints are satisfied.

Annotation Curation. We construct the *OmniRouting* benchmark using a scalable automated pipeline built on OmniSch (Lu et al. 2026) and OmniLayout (.). We collect 1,681 real-world schematic designs from open-source hardware platforms, including SparkFun, Arduino, Adafruit, GitHub, Seeed Studio, and ProtoCentral (SparkFun Electronics 2026; GitHub, Inc. 2026; Arduino 2026; Adafruit Industries 2026; Seeed Technology Co., Ltd. 2026; ProtoCentral 2026). While OmniLayout only supports PCB layout rendering for component placement, we develop a fully functional PCB EDA engine supporting both component placement and routing, capable of parsing source-level design information directly from EAGLE XML files, including component placements, board geometry, routing wires, vias, and copper polygons. Beyond annotation extraction, the engine also serves as an interactive visualization tool in our agentic evaluation framework (Section ??), rendering the current routing state as visual feedback for LMMs.

Statistics of OmniRouting Benchmark. As illustrated in Figure 3 and Figure 2, *OmniRouting* contains **1,681** real-world schematic-coupled PCB designs covering a broad range of application domains, including robotics, wireless systems, and sensing, providing a comprehensive benchmark for supervised learning and evaluation of PCB routing, and related EDA tasks. Our dataset includes three levels of structured annotations: (i) **Comprehensive Routing Task**, providing fine-grained annotations for **xx** components, **xx** pads, and **xx** nets, paired with aligned schematics and PCB layouts. Every PCB design is verified to be fully routable. Each component is annotated with its identity, value, placement, orientation, and layer assignment. (ii) **Reference Routing Annotations**, including an engineered verified reference routing consisting of routing **xx** wires, **xx** vias, and **xx** copper polygons, together with their geometric and manufacturing attributes (e.g., layer assignment, trace width, routing angle,

via properties, and polygon boundaries). These reference solutions enable both functional evaluation and reference-based routing similarity evaluation.

Evaluation Metrics

Evaluation Protocol

To the best of our knowledge, there is no standardized protocol for evaluating PCB routing predictions from LMMs that jointly considers connectivity correctness, design-rule compliance, electrical functionality, and agentic tool use for constraint-aware geometric reasoning. PCB routing is a well-known highly constrained combinatorial optimization problem with NP-hard substructures: given a set of placed components, pads, board boundaries, routing layers, obstacles, and a target netlist, the objective is to find feasible wire paths that connect electrically related terminals while avoiding overlaps, shorts, spacing violations, and other design-rule conflicts. The solution space grows combinatorially due to alternative routing paths, layer assignments, via placements, bend patterns, grid-level routing choices, and finite-width polygonal trace geometries, resulting in multiple functionally equivalent routing solutions with no known efficient algorithm guaranteed to find the global optimum. To evaluate a predicted routing design against a reference routing design, we formulate PCB routing evaluation as a graph comparison problem. Specifically, both the reference and predicted layouts are transformed into routing graphs (as shown in Figure 2), where nodes represent components, pads, and routing grid points, and edges represent physical routed connections. The evaluation then compares these graphs under electrical and geometric constraints, assessing whether the predicted routing preserves the intended net connectivity, avoids opens and shorts, satisfies spatial routing rules, and maintains the functional behavior of the reference design. **Given a set of placed components and pads V , target nets E , board boundary B , routing layers L , and obstacles $\mathcal{O}$, the routing quality is measured by:**

$$\min_{\mathcal{R}, \mathbf{z}, \mathbf{u}} W_r(\mathcal{R}) + \lambda_1 C_r(\mathcal{R}, \mathbf{z}, \mathbf{u}) + \lambda_2 D_r(\mathcal{R}, \mathbf{z}, \mathbf{u}, B, \mathcal{O}) + \lambda_3 \mathcal{F}_r(G^{\text{pred}}, G^{\text{ref}}) \quad (1)$$

subject to $r_e \subseteq B$, $z_e \in L$, and $u_e \in \mathcal{U}$, $\forall e \in E$, where $\mathcal{R} = \{r_e\}_{e \in E}$ denotes the set of predicted routing paths, z_e denotes the routing layer assignment, and u_e denotes the via configuration. $W_r(\mathcal{R})$ measures routing cost, including total

trace length, bend count, and via count, as a proxy for routing efficiency. $C_r(\mathcal{R}, \mathbf{z}, \mathbf{u})$ penalizes routing clearance violations, including wire-to-wire, wire-to-pad, wire-to-via, via-to-via, and pad-to-via violations. $D_r(\mathcal{R}, \mathbf{z}, \mathbf{u}, B, \mathcal{O})$ penalizes physical design-rule violations, including trace-width, boundary, keepout, obstacle-crossing, invalid-via, and layer-specific routing violations. $\mathcal{F}_r$ measures electrical functionality preservation between the predicted routing graph and the reference routing graph, defined as:

$$\mathcal{F}_r(G^{\text{pred}}, G^{\text{ref}}) = \alpha_1 \mathcal{D}_{\text{conn}}(G^{\text{pred}}, G^{\text{ref}}) + \alpha_2 \mathcal{D}_{\text{short}}(G^{\text{pred}}) + \alpha_3 \mathcal{D}_{\text{path}}(G^{\text{pred}}, G^{\text{ref}}). \quad (2)$$

where G^{pred} and G^{ref} are routing graphs extracted from the predicted and reference layouts, respectively. In each graph, nodes represent pads, pins, vias, junctions, and routing grid points, while edges represent physical routed copper segments. $\mathcal{D}_{\text{conn}}$ evaluates whether all terminals in each net are electrically connected, $\mathcal{D}_{\text{short}}$ penalizes unintended connections between different nets, and $\mathcal{D}_{\text{path}}$ measures graph-level routing discrepancy, including differences in path topology, layer transitions, via usage, and route geometry. The coefficients $\lambda_1, \lambda_2, \lambda_3$ and $\alpha_1, \alpha_2, \alpha_3$ balance the contribution of each evaluation dimension.

Experiment and Findings

Experimental Setups

Study Setup. The tested LMMs in this section include GPT-5.5 (OpenAI 2026), GPT-5-mini (?), Gemini-3.1-Pro-Preview (?), Gemini-2.5-Flash-Lite (?), Claude-Sonnet-4.6 (Anthropic 2026), LLaMA-4-Maverick (?), Mistral-Medium-3.5 (128B) (?), Qwen3.5-9B (?), and Qwen3.6-35B-A3B (Qwen Team 2026). We evaluate these models alongside two commercial and five open-source PCB placement baselines. We conduct two studies: (i) **Zero-shot Evaluation**, where all LMMs are prompted once without intermediate interaction to assess end-to-end PCB component placement. All settings include the board boundary, netlist, and component dimensions, while we further study the impact of additional information, including schematic images, schematic-aware netlist graphs, global schematic positions, and few-shot examples; and (ii) **Agentic Evaluation**, where LMMs are equipped with layout visualization, overlap checking, and routability verification tools to iteratively refine component placement under geometric and electrical constraints.

Implementation of Commercial and Open-Source PCB Layout Baselines. To comprehensively evaluate the difficulty and quality of our benchmark, we assess placement performance across both commercial and academic baselines. For commercial tool, we evaluate Quilter (Quilter 2026), a reinforcement learning-based PCB PnR solution. For academic baselines, we implement SA-PCB (?), an OpenROAD-based simulated annealing placer; NS-Place (?), a net separation-oriented analytical placement method; Cypress (?), a VLSI-inspired GPU-accelerated placement framework; Chen et al. (?), a clearance-constrained global placement method supporting heterogeneous components; and PCBAgent (?), an LLM and reinforcement learning-based placement framework.

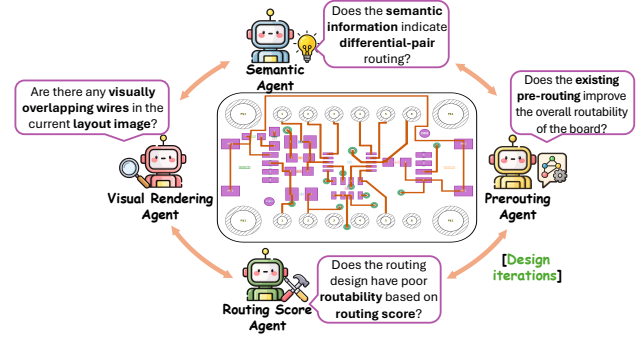


Figure 4: Overview of a multi-modal, multi-agent, multi-round agentic framework for evaluating LMM-based tool usage in PCB component placement.

Implementation of LMMs Agentic Framework. We design a multi-modal, multi-agent, and multi-round agentic evaluation framework where LMMs are provided with the board boundary, netlist, component dimensions, and schematic diagram, and equipped with three tools: layout visualization for rendering the current placement state, overlap checking for reporting constraint violations, and routability verification for estimating routing conflicts. At each step, the LMM observes the rendered layout and iteratively refines component positions, orientations, and layer assignments until all constraints are satisfied or the maximum number of steps is reached.

Main Results and Findings

Evaluation results of zero-shot study and agentic study are shown in Table ?? and Table 2. Overall, the dense component placement performance of LMMs remains significantly below that of human engineers, highlighting their limited capability in comprehensive layout design. Evaluation results reveal consistent limitations across multiple layout design capabilities.

- **Geometric Legality.** While LMMs demonstrate promising performance on simple PCB layouts with fewer than 40 components per design, their performance degrades significantly as complexity and component density increase. Furthermore, LMMs exhibit poor spatial utilization, placing 80% of components on the top layer rather than distributing them across available routing layers, suggesting a limited understanding of three-dimensional spatial organization in PCB design.

- **Routability.** LMMs exhibit substantial weakness in routability-aware placement, treating component arrangement as a purely spatial packing problem while neglecting circuit connectivity. Although zeroshot LMMs achieve promising score in overlap of less than 10% and out of boundary of less than 10%, their placements consistently result in poor routing outcomes reflected in higher PRWL and lower RR, suggesting a fundamental gap in understanding the relationship between component placement and downstream routing feasibility.

- **Electrical Functionality.** LMMs achieve an average similarity score below 60% when comparing predicted

Table 1: **One-shot evaluation of LMMs on PCB routing.** *Pass Rate* is the fraction of expected inputs that produced a loadable output. *NR* is the fraction of all required dataset nets that are fully connected and free of published design-rule violations. *PR* is the number of independent DRC-clean pad-to-pad connections divided by the minimum required count $\sum_n (|P_n| - 1)$; pad-to-via and via-to-via subdivisions do not add connections. Missing outputs remain in both denominators. *Open (%)*, *PShort (%)*, and *NShort (%)* are the percentages of all required dataset nets affected by at least one open, physical short, or logical short, respectively; each affected net is counted once. *Total*, *Clr.*, and *Dist.* are per-board means for total published violations, different-net copper clearance, and board-edge distance violations. *Polygon (%)* is the percentage of evaluated outputs containing at least one copper pour and is not a violation. *TWL*, *#Vias*, *#Layers*, and *RT* denote routed-wire centerline length, via count, used routing-layer count, and runtime. Higher is better for Pass Rate, NR, and PR; lower is better for all violation, length, runtime, and via metrics. “-” denotes unavailable results.

Model	Size	Experiment	Routability		Connectivity			Design Rules				Routing Quality			Efficiency	
			NR	PR	Open (%)	PShort (%)	NShort (%)	Total	Clr.	Dist.	Polygon (%)	TWL	#Vias	#Layers	Pass Rate	RT
Commercial LMMs																
GPT-5.5	-	Base	0.091	0.123	20.09	75.37	1.00	431.29	427.21	4.08	0.00	791.66	20.53	1.74	0.860	133.79
		Netclass detail	0.099	0.128	20.66	75.12	1.14	401.31	397.07	4.24	0.00	774.93	21.02	1.79	0.864	133.02
		Unlimited layer	0.090	0.117	31.46	71.24	0.92	367.08	362.50	4.58	0.00	781.98	22.79	4.91	0.854	130.22
		Fewshot	0.094	0.131	19.03	80.07	1.39	346.63	342.61	4.03	0.00	685.34	10.56	1.75	0.958	117.35
		Prerouter	0.100	0.142	22.38	77.95	1.51	354.44	349.52	4.92	0.00	735.99	27.25	1.88	0.900	127.46
GPT-5-mini	-	Base	0.058	0.067	31.20	75.49	3.11	372.07	368.80	3.28	0.00	648.43	15.25	1.38	0.989	69.89
		Netclass detail	0.056	0.060	38.91	68.96	4.58	294.29	291.18	3.11	0.00	552.48	8.95	1.27	0.993	62.52
		Unlimited layer	0.051	0.053	43.15	68.87	4.30	303.77	301.08	2.69	0.00	567.12	11.13	2.15	0.990	62.61
		Fewshot	0.041	0.044	61.91	53.70	10.18	150.67	148.81	1.87	0.00	323.94	3.11	1.26	0.991	55.87
		Prerouter	0.059	0.061	40.08	66.69	4.93	272.69	269.96	2.73	0.00	511.56	14.44	1.31	0.992	66.19
Claude Opus 4.8	-	Base	0.126	0.154	55.05	73.71	3.61	150.53	147.72	2.80	0.00	494.41	13.27	1.95	0.990	42.06
		Netclass detail	0.124	0.142	55.73	71.59	4.11	136.28	133.27	3.01	0.00	469.17	11.12	1.95	0.988	39.62
		Unlimited layer	0.121	0.134	59.06	69.82	3.31	124.32	121.41	2.91	0.00	468.88	15.35	4.63	0.989	39.99
		Fewshot	0.104	0.113	44.03	62.32	4.09	126.28	123.83	2.45	0.00	424.89	7.72	1.89	0.945	37.76
		Prerouter	0.125	0.144	53.13	72.44	4.35	141.07	137.74	3.33	0.00	480.48	15.77	1.95	0.991	41.43
Gemini 3.1 Pro-Preview	-	Base	0.007	0.009	1.71	3.45	0.05	157.63	155.64	1.99	0.00	428.92	20.82	1.89	0.060	260.00
		Netclass detail	0.004	0.005	1.12	2.47	0.05	183.78	178.71	5.07	0.00	442.01	20.16	1.92	0.046	307.41
		Unlimited layer	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Fewshot	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Prerouter	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Gemini 2.5 Flash-lite	-	Base	0.014	0.013	16.09	25.07	9.27	419.47	414.89	4.58	0.00	427.48	42.48	1.41	0.376	21.20
		Netclass detail	0.013	0.013	18.16	23.77	9.72	385.07	379.96	5.11	0.00	370.20	34.88	1.41	0.367	22.62
		Unlimited layer	0.011	0.010	18.16	23.57	9.50	363.90	358.39	5.50	0.00	361.86	31.00	1.39	0.362	21.67
		Fewshot	0.001	0.001	6.80	6.37	3.49	856.26	834.67	21.59	0.00	419.05	0.03	1.71	0.132	25.66
		Prerouter	0.008	0.007	15.40	18.74	7.82	439.41	433.26	6.16	0.00	370.76	37.42	1.50	0.315	27.05
Open-Source LMMs																
LLaMA 4 Maverick	400B	Base	0.043	0.051	28.70	55.76	26.75	235.45	231.39	4.07	0.00	420.04	11.34	1.74	0.834	20.18
		Netclass detail	0.034	0.038	33.63	55.58	34.47	238.21	234.30	3.91	0.00	382.55	11.29	1.69	0.839	25.13
		Unlimited layer	0.036	0.037	33.30	55.89	33.74	238.16	234.65	3.51	0.00	382.18	10.75	1.59	0.846	24.99
		Fewshot	0.004	0.006	21.37	22.38	14.57	152.29	145.79	6.50	0.00	219.58	3.32	1.61	0.510	33.16
		Prerouter	0.030	0.032	32.70	52.61	33.16	225.96	222.23	3.73	0.00	368.43	14.35	1.75	0.821	25.91
Mistral Large 3	-	Base	0.037	0.042	31.45	52.32	21.57	257.39	252.77	4.62	0.00	432.39	9.16	1.95	0.835	84.54
		Netclass detail	0.037	0.039	30.38	54.34	23.26	276.19	272.08	4.10	0.00	438.33	8.57	1.87	0.842	86.26
		Unlimited layer	0.037	0.039	30.70	56.57	24.70	290.65	286.56	4.09	0.00	452.39	6.25	1.58	0.853	86.75
		Fewshot	0.031	0.033	25.50	43.42	18.80	225.35	221.55	3.80	0.00	361.71	1.31	1.54	0.782	102.77
		Prerouter	0.036	0.039	33.36	57.63	24.33	283.54	278.88	4.67	0.00	447.21	19.13	1.91	0.856	89.05
Qwen3.5-9B	9B	Base	0.012	0.016	62.69	20.45	6.48	69.77	63.86	5.91	0.00	131.34	0.09	1.01	0.725	-
		Netclass detail	0.011	0.024	60.58	25.37	5.00	78.15	74.33	3.83	0.00	161.32	0.09	1.03	0.725	-
		Unlimited layer	0.013	0.024	60.81	24.48	4.93	73.28	69.29	3.99	0.00	158.99	0.08	1.02	0.730	-
		Fewshot	0.011	0.011	27.31	12.44	3.94	128.02	124.18	3.84	0.00	185.32	0.00	1.03	0.444	-
		Prerouter	0.011	0.017	54.98	23.43	4.97	90.30	85.76	4.55	0.00	163.74	0.41	1.06	0.656	-
Qwen3-235B-A22B	235B-A22B	Base	0.052	0.071	32.50	78.25	21.21	375.97	371.62	4.35	0.00	611.40	5.82	1.52	0.991	-
		Netclass detail	0.042	0.057	33.53	64.74	17.36	331.31	327.14	4.16	0.00	540.49	5.38	1.46	0.942	-
		Unlimited layer	0.044	0.057	32.41	63.75	17.93	341.12	335.54	5.58	0.00	540.17	5.13	1.44	0.932	-
		Fewshot	0.029	0.031	26.06	39.65	19.63	357.20	350.44	6.76	0.00	429.82	1.41	1.43	0.749	-
		Prerouter	0.036	0.045	35.36	53.93	16.55	270.98	266.62	4.35	0.00	443.68	7.07	1.50	0.910	-

placements against reference layout designs, indicating that component positions deviate substantially from both the schematic diagram’s visual topology and the expert-designed reference in terms of relative spatial distances. This quantitative gap reflects a fundamental deficiency: LMMs lack a structured understanding of circuit hierarchy and functional grouping, and consequently fail to preserve the spatial relationships of subcircuits during placement.

• **Ablation Study Analysis.** We conduct an ablation study examining the contribution of three input modalities:

schematic diagram, netlist with spatial weights, and semantic component attributes. Among these, fewshot yields the greatest performance improvement, highlighting its importance in guiding layout reasoning. Additionally, few-shot prompting consistently yields significant performance gains across models, demonstrating the benefit of in-context layout examples.

• **Agentic Spatial & Routing Reasoning.** By incorporating layout visualization, geometric legality checking, and routability feedback into a multi-agent framework, agentic in-

Table 2: **Agentic evaluation of LMMs on PCB routing.** *Pass Rate* is the fraction of the 500 expected inputs that produced a loadable output. *NR* is the fraction of all required dataset nets that are fully connected and free of published design-rule violations. *PR* is the number of independent DRC-clean pad-to-pad connections divided by the minimum required count $\sum_n (|P_n| - 1)$; pad-to-via and via-to-via subdivisions do not add connections. Missing outputs remain in both denominators. *Open (%)*, *PShort (%)*, and *NShort (%)* are the percentages of all required dataset nets affected by at least one open, physical short, or logical short, respectively; each affected net is counted once. *Total*, *Clr.*, and *Dist.* are per-board means for total published violations, different-net copper clearance, and board-edge distance violations. *Polygon (%)* is the percentage of evaluated outputs containing at least one copper pour and is not a violation. *TWL*, *#Vias*, *#Layers*, *RT*, and *Avg. Steps* denote routed-wire centerline length, via count, used routing-layer count, runtime, and recorded agent iterations. Higher is better for Pass Rate, NR, and PR; lower is better for violations, length, runtime, vias, and steps. “-” denotes unavailable results.

Model	Size	Agent Configuration	Routability		Connectivity			Design Rules			Routing Quality			Efficiency			
			NR	PR	Open (%)	PShort (%)	NShort (%)	Total	Clr.	Dist.	Polygon (%)	TWL	#Vias	#Layers	Pass Rate	RT	Avg. Steps
Commercial LMMs																	
GPT-5.5	-	No Tools	0.217	0.248	55.96	39.53	0.39	48.06	42.87	5.19	0.00	578.95	29.11	1.97	1.000	1060.60	16.97
		Visualization	0.210	0.245	56.89	36.99	0.39	37.81	32.90	4.91	0.00	549.45	26.34	1.97	1.000	1139.14	17.09
		Routing Score	0.248	0.210	58.10	10.07	0.27	9.59	5.58	4.01	0.00	332.51	15.50	1.96	1.000	779.40	20.26
		Semantic Input	0.218	0.256	55.39	38.24	0.62	41.91	36.82	5.09	0.00	554.48	28.10	1.96	1.000	1037.56	16.58
		Prerouting/Remake All Tools	0.226	0.303	49.74	43.11	0.42	63.43	57.92	5.51	0.00	606.21	30.81	1.98	0.996	988.91	15.80
			0.280	0.256	54.66	9.12	0.26	9.35	5.27	4.08	0.00	326.57	17.88	1.96	0.998	737.76	20.78
GPT-5-mini	-	No Tools	0.099	0.076	53.33	56.91	8.10	91.75	88.98	2.77	0.00	289.96	11.16	1.78	1.000	368.76	18.38
		Visualization	0.099	0.074	54.50	53.31	7.34	80.85	78.95	1.91	0.00	247.67	8.61	1.75	1.000	385.25	20.00
		Routing Score	0.099	0.064	67.40	29.86	2.22	21.16	19.92	1.24	0.00	113.28	4.56	1.75	0.998	355.64	23.88
		Semantic Input	0.091	0.072	52.70	54.24	7.76	89.13	86.92	2.21	0.00	255.54	6.86	1.69	0.992	413.85	19.42
		Prerouting/Remake All Tools	0.092	0.105	35.89	43.89	5.35	77.59	74.60	2.99	0.00	309.55	15.49	1.92	0.892	360.66	17.66
			0.068	0.057	55.91	24.57	2.02	22.57	20.77	1.80	0.00	118.84	7.49	1.74	0.914	410.83	23.65
Claude Sonnet 4.6	-	No Tools	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Visualization	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Routing Score	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Semantic Input	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Prerouting/Remake All Tools	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Gemini 3.1 Pro-Preview	-	No Tools	0.022	0.031	10.06	10.47	0.17	116.24	111.00	5.24	0.00	897.01	35.19	1.95	0.188	1312.46	6.40
		Visualization	0.018	0.026	9.27	8.69	0.06	111.13	106.18	4.95	1.28	925.45	36.41	1.96	0.156	1692.32	12.09
		Routing Score	0.004	0.004	1.71	1.36	0.00	45.72	41.61	4.11	0.00	448.71	24.61	1.94	0.036	2034.23	14.39
		Semantic Input	0.026	0.038	12.36	15.54	0.40	135.57	131.56	4.01	4.10	864.10	32.11	1.92	0.244	1198.73	6.05
		Prerouting/Remake	0.041	0.053	11.91	14.12	0.14	109.10	105.89	3.20	2.21	725.41	30.71	1.96	0.272	1131.42	6.42
		All Tools	0.012	0.010	4.65	3.14	0.15	55.07	51.63	3.44	0.00	551.93	22.27	1.76	0.090	1876.04	15.40
Gemini 2.5 Flash-lite	-	No Tools	0.003	0.002	44.97	8.88	4.55	216.17	215.06	1.11	0.00	98.94	2.58	1.10	0.540	65.48	22.51
		Visualization	0.001	0.001	48.55	8.39	4.62	111.09	110.52	0.57	0.00	61.98	0.27	0.86	0.620	92.98	23.34
		Routing Score	0.002	0.002	63.35	12.45	4.97	30.44	29.39	1.05	0.00	40.20	0.99	1.03	0.740	64.33	24.71
		Semantic Input	0.001	0.001	51.96	5.37	2.62	58.78	55.76	3.02	0.00	40.81	0.38	0.40	0.658	35.78	24.36
		Prerouting/Remake	0.005	0.011	15.56	13.37	8.14	255.80	249.48	6.33	0.00	222.90	14.61	1.62	0.342	79.62	14.63
		All Tools	0.005	0.007	32.16	12.37	5.07	80.73	76.97	3.76	0.00	101.73	8.81	1.38	0.494	59.99	23.20
Open-Source LMMs																	
LLaMa 4 Maverick	400B	No Tools	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Visualization	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Routing Score	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Semantic Input	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Prerouting/Remake All Tools	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Ministral 14B	14B	No Tools	0.004	0.007	68.31	43.24	23.94	230.99	226.89	4.10	0.22	230.00	4.40	1.58	0.896	399.42	9.60
		Visualization	0.005	0.008	77.99	37.75	18.75	130.51	127.31	3.20	0.42	161.60	2.10	1.39	0.960	296.62	9.71
		Routing Score	0.008	0.008	81.70	26.63	10.04	34.13	31.97	2.15	0.00	85.78	1.25	1.47	0.972	388.79	9.97
		Semantic Input	0.003	0.005	50.18	29.91	15.70	205.97	203.78	2.20	1.06	177.15	2.03	1.30	0.754	352.02	9.67
		Prerouting/Remake	0.008	0.027	56.63	43.45	24.30	234.28	230.20	4.09	0.00	258.26	14.73	1.82	0.840	356.41	9.34
		All Tools	0.010	0.022	53.01	18.12	6.54	28.53	27.77	0.77	0.00	92.23	4.45	1.62	0.818	345.07	10.00
Qwen3.5-9B	9B	No Tools	0.014	0.009	19.84	15.71	6.75	130.87	129.93	0.94	0.76	264.21	1.47	1.16	0.262	477.12	18.81
		Visualization	0.018	0.010	24.82	17.57	6.82	98.55	97.20	1.35	0.00	226.80	1.06	1.19	0.330	436.80	20.53
		Routing Score	0.014	0.008	21.48	14.80	4.78	79.42	78.95	0.48	0.00	200.14	1.17	1.35	0.266	506.80	21.29
		Semantic Input	0.001	0.001	2.66	1.75	0.43	81.18	80.95	0.23	0.00	144.77	3.55	1.45	0.044	241.65	4.86
		Prerouting/Remake	0.001	0.003	3.43	4.17	2.08	332.02	331.86	0.16	0.00	371.58	5.27	1.66	0.088	322.00	10.80
		All Tools	0.001	0.001	2.56	1.70	0.72	102.32	102.09	0.23	0.00	178.20	5.55	1.77	0.044	149.93	4.95
Qwen3-235B-A22B	235B-A22B	No Tools	0.000	0.000	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.002	63.67	1.00
		Visualization	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Routing Score	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Semantic Input	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
		Prerouting/Remake All Tools	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

teraction improves placement performance by reducing component overlap and out-of-boundary violations, while layout visualization feedback contributes to improved routing performance compared to direct generation baselines.

• **Spatial Reasoning Steps.** More iterations lead to improved placement performance, as models have greater opportunity to resolve constraint violations. Gemini 3.1 Pro-Preview models converge rapidly, saturating performance around 5 iterations, whereas Qwen models fail to converge

even at the maximum limit of 50 iterations, suggesting fundamental differences in spatial reasoning efficiency and constraint satisfaction across model families. Notably, Qwen3.5-9B and Qwen3.6-35B-A3B fail to produce structured outputs entirely.

Conclusion

We introduce OmniRouting, a benchmark for evaluating LMMs on PCB routing geometric and topological reason-

ing, consisting of 1,681 real-world PCB routing tasks coupled with fine-grained, engineer-verified reference routing annotations and structured evaluation protocols. We evaluate state-of-the-art LMMs under both end-to-end and tool-augmented settings, alongside open-source and commercial PCB routing tools as baselines. Our analysis reveals critical limitations of LMMs, including weak multi-net routing reasoning, poor understanding of layer and via assignment, limited congestion and routability optimization, and routing solutions that exhibit deviations from reference designs across connectivity preservation, routing topology, trace geometry, layer assignment, and design-rule compliance.

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